

VR Interface vs Desktop to convey Quality of Outerwear: a comparative study

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ABSTRACT

Conveying the quality of garments through media as in the physical store is demanding. This study proposes the design of a VR interface that aims to convey outerwear quality, featuring the 3D model animated through physical simulation. To measure the effectiveness of this interface, it is tested simultaneously with its corresponding desktop counterpart, with the photo gallery of the product, in a within-subject analysis on 50 users. Results show that perceived quality of products changes between the experiences. Moreover, in VR visual content was found to be more significant for the quality assessment than written information.

Keywords: Virtual reality, user interface, quality, perception, outerwear.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—User interface design; Human-centered computing—Interaction design—Empirical studies in interaction design

1 INTRODUCTION

Digitalization of clothing stores is challenging because of its importance for the assessment of garment quality. The object of this research is the value that VR can bring to the digital shopping experience for fashion products. If companies exploited new media to better communicate product information and fit user's expectation with reality, there would be fewer misguided purchases. Therefore, the research question driving this study is as follows.

RQ: Does the perception of outerwear vary according to the media and the visual content used in its communication?

2 RELATED WORKS

To convey the quality of outerwear it is necessary to comprehend its meaning. For this reason, literature related to quality theory has been analyzed.

Quality Perception The first distinction that occurs when analyzing quality is between objective quality and perceived quality. The objective quality of a garment is a result of material features [1, 2], which determines the durability. For this reason, we analyze empirical studies that record the reasons for disposal of garments to understand what are the features that the user should analyze when selecting a product by its quality [4]. Specifically, the items that could affect the outerwear are: hole or tear, looks very used or worn, stains, worn out, color change or fading, lost elasticity, broken seam or sewing failure, broken zipper, material has become thin, washed out, failed mending or repair, print faded, threads drawn

out, broken decoration, felting, buttons missing [4]. In addition to these parameters, according to interviewed outerwear manufacturers, the padding is important in assessing the quality of a coat, as if it is of different constitution, it behaves differently to stresses and differently preserving the original shape of the product. Technical qualities can be influenced by user goals and needs, as well as being subject to user perception and interpretation [5]. In fact, users are not aware of the objective quality, but earn the perceived quality through senses [3]. Quality cues are responsible for the formation of perceived quality prior to consumption and are divided into intrinsic cues and extrinsic cues [1]. Intrinsic cues cannot be changed without changing the product itself, they are part of it, they correspond to technical properties, and an example of them are material features. Extrinsic cues are external aspect that are related or associated to the product, like brand, reputation, price or country of origin. This work specifically focuses on intrinsic cues since their link to the objective quality.

3 METHOD

In this study a VR interface for conveying quality of outerwear is designed. It is also developed its corresponding desktop version to carry a comparative study, to highlight the impact the choice of the media and the kind of visual content have in the communication effectiveness. Both interfaces contain the same products (4 female outerwear) with labels about intrinsic quality cues recognized as significant in literature. They are shape, color, materials, components (details such as zip, buttons, sewing, etc.), waterproofing, warmth, lightness, softness, smoothness, elasticity, thinness, sustainability. The source of the assets used (pictures and 3D models of products) is an Italian high-quality outerwear company.

3.1 Experiment design

System Development The desktop system is realized with Figma and displayed on laptop. The interface includes the photo-gallery of 3 picture of the product (front, ¾, back) on the left and the label on the right. On both sides there are commands to switch product.

The VR system is realized with Unity and shown with a Meta Quest 2 headset. Hand tracking enables interaction with elements in the scene. The interface contains the animated outerwear 3D model centered in front of the user, with the label on the right. The model is life-sized and can be rotated by sliding the hand along a horizontal slide bar. It is animated through physical simulation run in Clo3D on a walking female avatar. The model is simulated both with the zip closed and opened, so in the system a command to switch to the open outerwear could be implemented. On the sides there are commands to switch products, but in the form of small 3D models.

Experiment Stages A within-subject study was conducted to validate the following hypothesis:

H1: Visual content of the outerwear is more important than the written information for the assessment.

H2: The garment assessed as being the best quality can change according to media (desktop or VR) and visual content type (photo-gallery or 3D model in motion).

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Figure 1: A - Desktop interface developed with Figma. B - VR interface developed with Unity

H3: Each visualization highlights different quality cues of the outerwear.

The sample is made by 50 Italian women (mean age 39.7, sd 16.3), that have some previews experience of online purchasing of clothes and have no expertise with virtual reality. To conduct the survey a custom questionnaire is prepared, given the peculiarity of the hypothesis.

The structure of the experiment starts with a demographic questionnaire. Then the first interface is shown to the user and is asked to inspect and compare the garments with the aim of understanding which of them is of the best quality. The user is unaware that there is no right answer as all products are high-quality. The user is assisted during the experiment in case of need or doubts about the commands. Then the user fills in the evaluation form related to the experience. The desktop and VR experiences follow the latin square experiment design.

The evaluation form asks about the usability of the interface and the vividness of the visual content, to validate the solidity of the systems. It is also asked if more visual content or label has been observed to assess quality. Then the user tells his preference about the product being of the best quality. Finally, in a 5-point Likert scale it is asked for each quality cue in what extent the user was able to evaluate it.

4 RESULTS

About the importance of the visual content over the written information, data are heterogenous, therefore there is high subjectivity in quality assessment by users. Nevertheless, while in the desktop experience there is no predominancy, in the VR experience the visual content is observed more than the label. That means in VR users focus more on the visual content, validating H1.

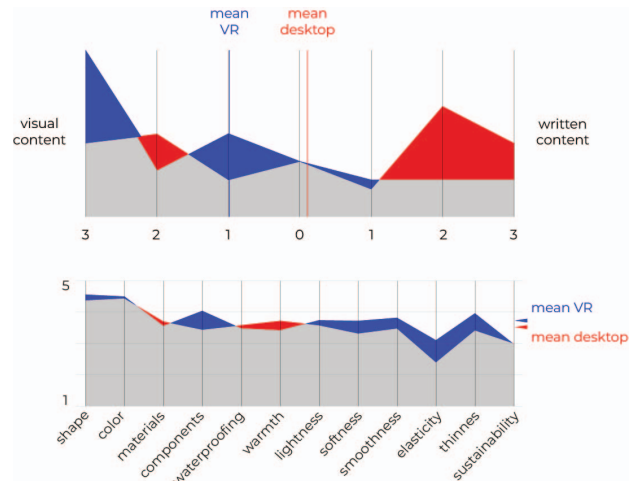


Figure 2: Blue = VR. Red = desktop. A - Likert scale ratings between visual content and written content. B - 5-Point Likert scale mean rates of quality cues perceptibility.

Considering the product declared to be of the best quality, it changes between desktop and VR experiences for 54% of the users. None of the products presents high dominance in rates, that means visual contents quality is consistent and does not affect results. That means each kind of visualization can highlight products differently, therefore validating H2.

Finally, VR interface conveys slightly better some quality cues of the outerwear. By looking at each single cue, warmth is perceived slightly better in desktop, while components (zippers, buttons, sewing, etc.), softness, smoothness, elasticity, and thinness are perceived better in VR. Most of the cues enhanced by VR result from the visual content quality as well as the impact that animation has on conveying physical properties. Despite the higher perception of quality cues in the VR experience, the difference is not wide, and therefore H3 is partially validated.

5 CONCLUSIONS

The choice of the media and type of visual content can play a key role in conveying garment quality. Virtual reality can improve quality assessment of garments making purchase behavior more responsible.

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